Backing Up VM Files with Borg

BorgBackup, or Borg, is a de-duplicating open source program that also offers compression and encryption. The main goal of Borg is to provide efficient and secure storage of backed up data. In this article, the author shows how Borg can securely back up a VM, and has added some of his own script modifications to make it more efficient.

I have been trying to back up VMs with various backup tools. This turned out to be expensive for the volume of data that seems to be growing larger every day. It was becoming important for me to back up data and at the same time reduce the cost of storage, which is calculated per GB. If the data is to be stored as per HIPPA regulations, the costs get even higher. Backing up VM files was a challenge, particularly when done live or in real-time.

As a consultant, I had to deliver a cost-effective tool to back up data and store it in an encrypted format.

The client was advised to use KVM on CentOS instead of VMware and Windows. This first option proved successful and the performance was better than the VMware-Windows combination. It was necessary to use a backup tool to store the data in a safe place even when at rest, as it contained PMI data. I scoured Google for all the possible tools prior to deciding on one -- and the options were either beyond budget or did not deliver what was needed, i.e., space, efficient storage, speed, data encryption compression, robustness, being time tested, reliable, and easy to use.

I eventually landed at the Borg site (*https://borgbackup.readthedocs.io)*, but I was sceptical about it as no one had mentioned it, nor had I heard of it earlier. But I read through the site, found it very interesting, and decided to try out Borg.

It turned out to perfectly match my needs, as it had:

- Space efficient storage
- Speed
- Data encryption
- Compression
- Off-site backups
- Backups mountable as file systems

And it was free and open source software.

Borg is rather easy to install on multiple platforms and offers single-file binaries that do not require anything to be installed — you can just run them on the following platforms:

- Linux
- Mac OS X
- FreeBSD
- OpenBSD and NetBSD (no xattrs/ACLs support or binaries yet)
- Cygwin (experimental, no binaries yet)
- Linux sub-system of Windows 10 (experimental)

Installation

Borg x86/x64 AMD/Intel compatible binaries (generated with PyInstaller) are available on the release pages for Linux: *glibc >= 2.13* (okay for most supported Linux releases). Older *glibc* releases are untested and may not work.